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CASE STUDY

Delivery Performance Analysis

For a Logistics Company



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Understanding the Problem

A logistics company evaluates its delivery performance across regions, products, and time periods.

Delivery Delays

Significant gap between expected and actual delivery dates causing customer dissatisfaction.

Underperforming Partners

Certain delivery partners consistently underperform relative to others.

Operational Inefficiencies

Customer complaints highlight areas requiring process improvement.

Regional & Product Delays

Certain products or regions experience more frequent delays than others.

Objectives of the Case Study

01

Measure Delivery Performance

Quantify on-time vs delayed delivery rates across all regions.

02

Identify Underperforming Partners

Benchmark delivery partners and flag those with consistent delays.

03

Detect Product/Region-wise Delays

Pinpoint which products and regions experience the most delays.

04

Examine Monthly Performance Trends

Analyze delivery trends over time to detect seasonal patterns.

Data Collection

Data was collected from internal company sources:

Delivery Database

Historical order and shipment records with expected vs. actual delivery dates.

Order Management System

Real-time order tracking data including product type, region, and partner assignment.

Customer Feedback Records

Complaint logs and satisfaction ratings tied to specific deliveries and partners.

Data Cleaning & Exploration

1

Remove Duplicates & Handle Missing Values

Clean data by removing duplicate orders and addressing missing delivery dates.

2

Correct Inconsistencies & Standardize Formats

Fix inconsistent entries (product names, regions) and standardize date formats.

3

Feature Engineering

Create new columns like Delivery_Days and Delay; categorize: On-time / Slightly Late / Late.

4

Data Exploration

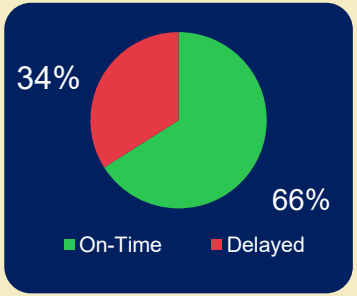
Analyze key metrics, identify outliers, and summarize performance by region, product, and partner.

Delivery Performance Dashboard

On-Time Deliveries	Delayed Deliveries
65–68%	32–35%

South region has the highest delays, especially for **Fragile & Furniture items.** On-time rates show a slight **declining trend** month-over-month, peaking during seasonal demand.

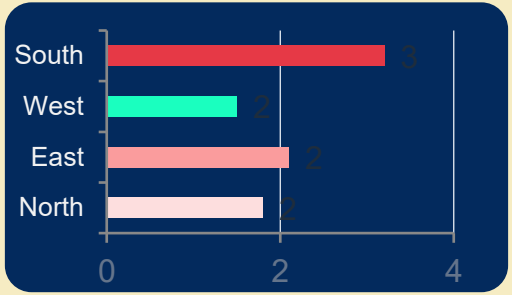
Overall delivery status



Delivery Partner Average Delay (Days)



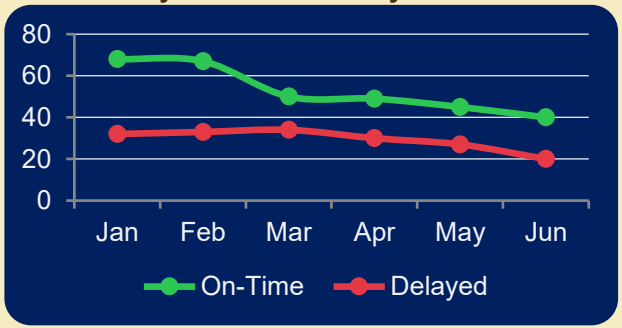
Region-wise Delays



Average Delay by Region & Product

	Electronics	*Clothing*	*Furniture*	*Fragile Items*
North	1.0	2.5	3.1	4.0
South	2.8	3.8	4.1	5.2
East	1.5	2.0	2.6	3.9
West	0.8	1.7	2.3	3.5

Monthly On-Time vs Delayed Deliveries



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DESCRIPTIVE ANALYTICS

- 65–68% of deliveries are on-time; 32–35% are delayed
- Average delivery delay: 2.2 days across all partners
- Partner A: 1.2d | Partner B: 3.4d (highest) | Partner C: 2.0d avg delay
- South region records the highest average delivery delays overall
- Fragile items is the most delayed product with an average of 5.1 days in the South region, followed by Furniture at 4.2 days
- From March to June there is a seasonal drop in on-time delivery performance

66%	34%	2.2 days
On-Time Rate	Delayed Rate	Avg Delay

Descriptive statistics summarize past delivery data to understand current performance baselines.

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DIAGNOSTIC ANALYTICS

- 32–35% delays → Poor route planning and no real-time tracking
- 2.2 days avg delay → Partner B's inefficiency inflates the overall average
- Partner B worst (3.4d) → Understaffed depots, weak last-mile routing
- South region highest delays → Fewer hubs, longer last-mile, fragile item concentration
- Fragile (5.1d) & Furniture (4.2d) worst → Special handling gaps and untrained staff at dispatch
- March–June drop → Seasonal surge with no capacity scale-up

Partner Performance
Root Cause #1

Region & Product Type
Root Cause #2

Seasonal Demand
Root Cause #3

Diagnostic analytics investigates patterns and correlations to explain why delivery failures occur.

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PREDICTIVE ANALYTICS

- Orders assigned to Partner B have 70%+ chance of delay; flagged pre-dispatch
- South region deliveries predicted high-risk, especially fragile and furniture items
- March–June period predicted to worsen; delay rate likely to cross 40% without intervention
- Fragile items in South projected to hit 6+ days average if handling gaps persist
- Partner B delay trend rising; projected to reach 4.0d average by Q3
- Current trend projects on-time rate dropping below 60% by end of season

Regression Model	Time Series	Classification
Delay Prediction	Trend Forecasting	Risk Flagging

Predictive analytics uses statistical models to forecast future delivery risks and trends.

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PRESCRIPTIVE ANALYTICS

- Partner B → Cap high-value orders; enforce SLA or reassign to Partner A
- South region → Open micro-hub; prioritise Partner A for fragile & furniture
- March–June → Pre-scale fleet and staff by February; set surge capacity targets
- Fragile items → Mandate certified handling staff and dedicated packaging lane
- Delay alert system → Auto-flag orders >2d risk at booking stage
- On-time KPI → Set 75% floor; trigger escalation if weekly rate drops below

Partner Routing
Optimization

Automated Alerts
Risk Management

KPI Dashboard
Continuous Monitoring

Prescriptive analytics recommends specific actions to optimize delivery performance outcomes.

Strategic Decisions

Optimize Delivery Operations

- Streamline processes to reduce 32–35% delay rate
- Deploy real-time alert system to auto-flag orders at delay risk before dispatch

Improve Partner Performance

- Enforce SLA with Partner B or reallocate to Partner A to bring avg delay below 2.0 days
- Redistribute high-value and fragile orders exclusively to Partner A until Partner B meets SLA

Product & Region-Specific Fixes

- Introduce product-specific SLAs — fragile items capped at 3.0 days, furniture at 3.5 days
- Open a micro-hub in South region with certified handlers for fragile and furniture items

Address Seasonal Trends

- Pre-scale fleet and workforce every February ahead of the March–June seasonal surge
- Adjust routes and inventory based on historical data

Continuous Monitoring

- Place Partner B on performance improvement plan with monthly delay targets
- Regular customer feedback loops to detect issues early

Conclusion

The analysis revealed significant delivery delays, underperforming partners, and region/product-specific challenges. By optimizing operations, focusing on high-risk areas, and monitoring performance trends, the company can improve delivery efficiency and customer satisfaction.

Operational Efficiency	Partner Optimization	Customer Satisfaction
66%	**5.2d**	**South**
On-Time Rate	Max Delay	Highest Delay Region

THANK YOU

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Data Analysis Case Study | Delivery Performance